

Probabilities

Ex ①: A student has to learn 20 subjects for his exam. But he learns only 15. The exam consists of 3 questions.

a. What is the probability that the student answers to 2 from 3 questions? $\rightarrow$ the 2 questions are from the 15 subjects $\rightarrow$

b. What is the probability that the student answers to at least 2 questions?

a. $P = \frac{\text{no. of favourable cases}}{\text{no. of total cases}} = \frac{\binom{15}{2} \cdot \binom{5}{1}}{\binom{20}{3}}$

$\rightarrow$ we have:

2 questions from 15 subjects (learned)

1 question from the remaining 5 subjects

$$= {}^{20}C_3$$

b. At least 2 questions $\Rightarrow$ 2 OR 3 questions $\Rightarrow$ Addition

$$P = \frac{\binom{15}{2} \cdot \binom{5}{1}}{\binom{20}{3}} + \text{(OR)} \frac{\binom{15}{3} \cdot \binom{5}{0}}{\binom{20}{3}}$$

the probability to answer to 2 questions

the probability to answer to 3 questions

Ex ② There are 20 candidates competing in a contest.

How many possibilities are to choose the winner, first runner up and second runner up?

Because the order matters $\Rightarrow$ we use arrangements

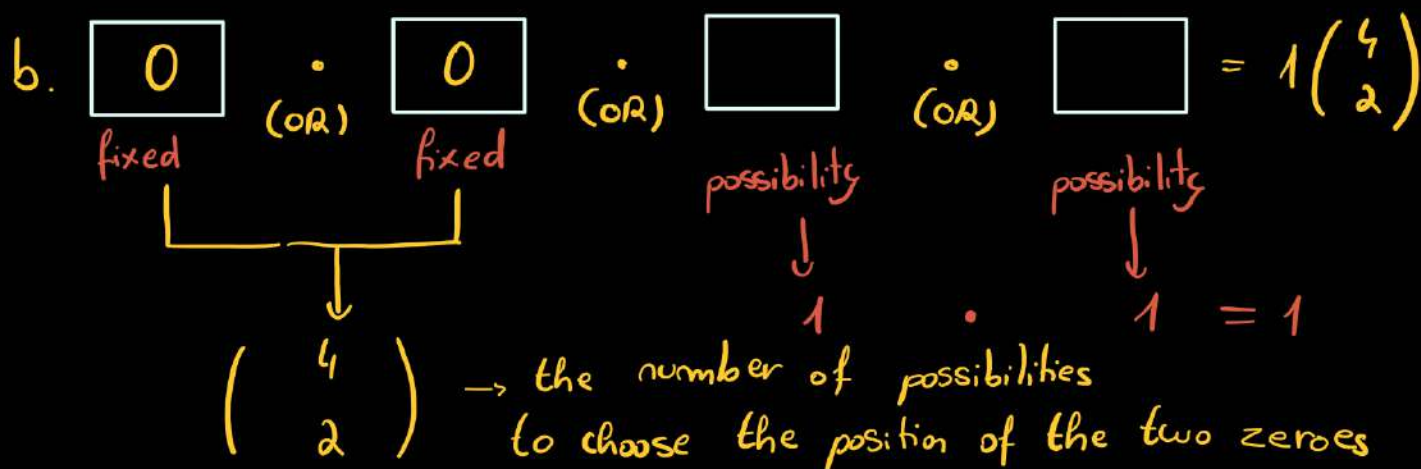
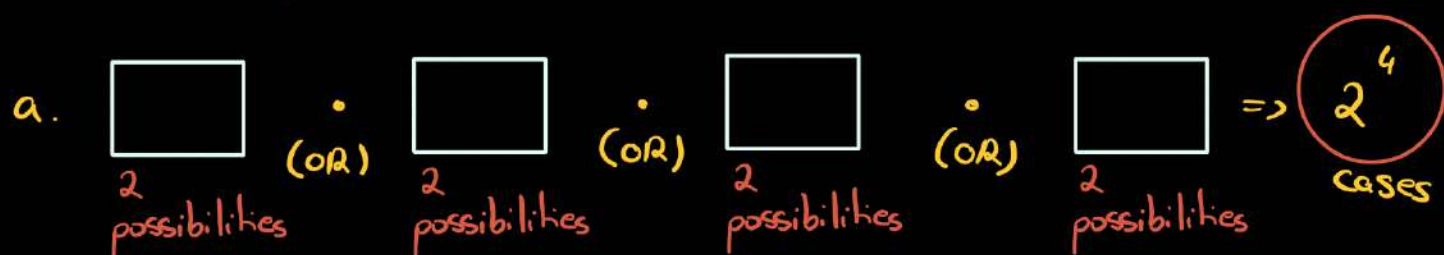


Ex ③ Consider a binary code with 4 bits, 0 (zero) or 1.

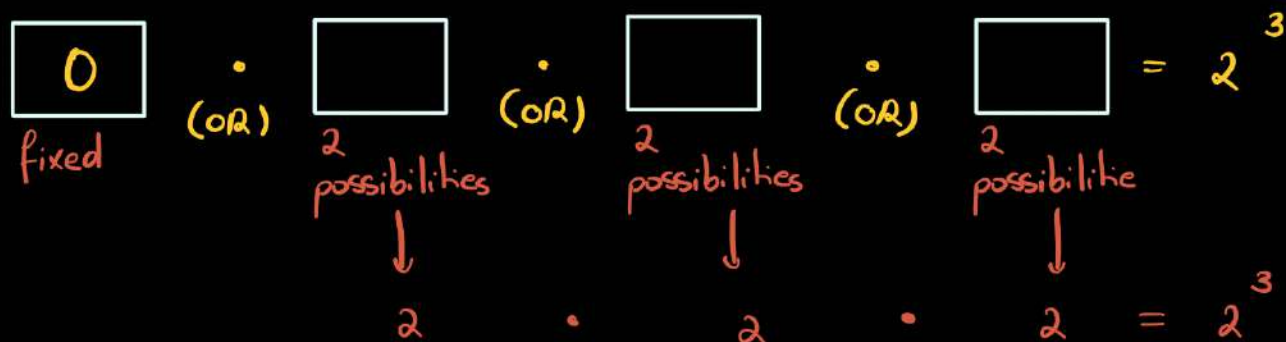
a. How many code words are?

b. How many code words have exactly two zeroes?

c. How many code words begin with a zero?



C.



Ex (4) Shuffle a deck of 52 cards, flip a coin and roll a dice.
Find the probability to pick a 10 (from the deck), a head (from the coin) and an even number (from the dice) at the same time.

$$P = \frac{4}{52} \cdot \frac{1}{2} \cdot \frac{3}{6} \rightarrow \text{there are three even numbers on a dice with six numbers in total}$$

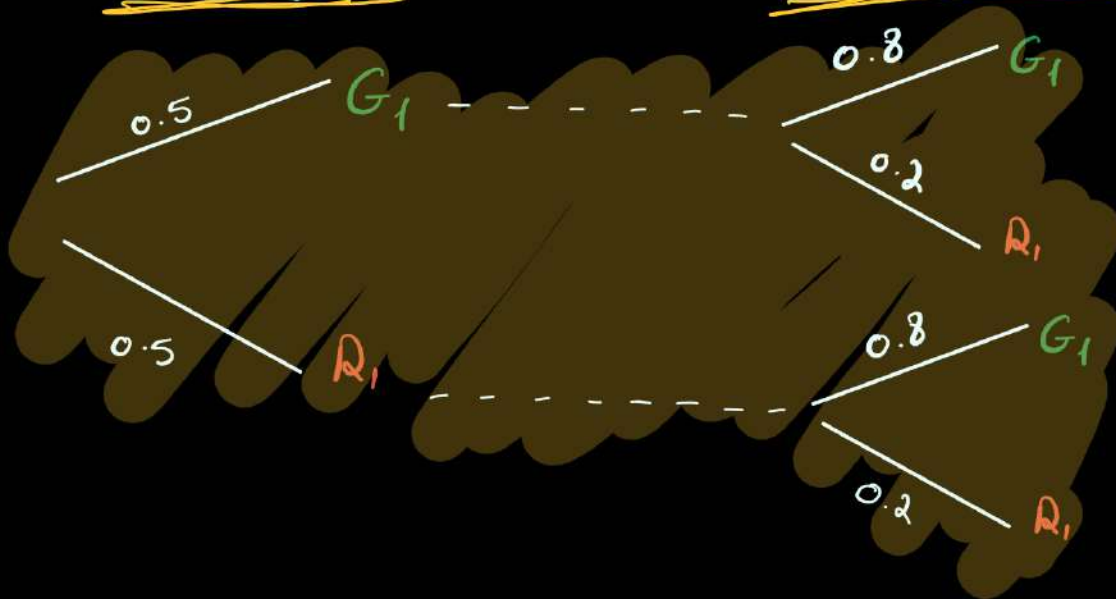
$\downarrow$ the possibility to pick a ten out of the four tens from the deck
 $\downarrow$ 50% chance for the coin to show the head side

Ex (5) Two traffic lights. At arrival, the second light has the same color as the first one with probability 0.8, and we assume that the first light is equally likely to be red or green.

- What is the probability that the second light is green?
- What is the probability that you wait for at least one light?
- What is the probability that the first light is green given a red second light?

First light

Second light



Tree
Diagram

$$a. G_2 = G_1 G_2 \cup_{(OR)} R_1 G_2$$

$$P(G_2) = P(G_1 G_2) + P(R_1 G_2) = 0.5 \cdot 0.8 + 0.5 \cdot 0.2 = 0.4 + 0.1 = \underline{\underline{0.5}}$$

b. $\Rightarrow$ at least one light is RED

$$X = G_1 R_2 \cup_{(OR)} R_1 G_2 \cup_{(OR)} R_1 R_2$$

OR

$$\bar{X} = G_1 G_2$$

$\Rightarrow$

$$P(G_1 R_2) + P(R_1 G_2) + P(R_1 R_2)$$

$$\Rightarrow P(X) = 1 - P(\bar{X}) = 1 - P(G_1 G_2) = 1 - 0.5 \cdot 0.8 = 0.6$$

C. GREEN first, RED second

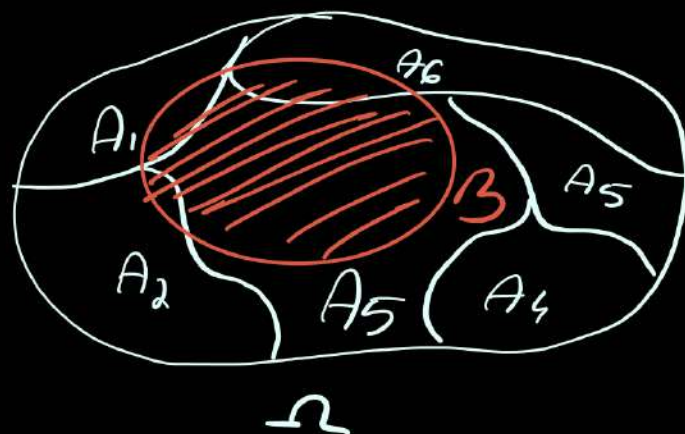
$$P_B(A) = \frac{P(A \cap B)}{P(B)} \rightarrow \text{the def of conditional probability}$$

$$\Rightarrow P_{R_2}(G_1) = \frac{P(G_1 R_2)}{P(R_2)} = \frac{0,5 \cdot 0,2}{P(G_1 R_2) + P(R_1 R_2)} = \frac{0,5 \cdot 0,2}{0,5 \cdot 0,2 + 0,5 \cdot 0,8}$$

Law of Total Probabilities Baye's Formula

if we have a Space Event: $\{A_1, A_2, \dots, A_n\}$
 $\Rightarrow$ the union covers the hole Space
 $\Rightarrow$ the intersection is the empty set

$$P(B) = P(A_1) \cdot P_{A_1}(B) + P(A_2) \cdot P_{A_2}(B) + \dots + P(A_n) \cdot P_{A_n}(B)$$



Bayes' Formula:

$$P_A(B) = \frac{P_B(A) \cdot P(B)}{P(A)} \quad \Bigg| \quad P_B(A) = \frac{P_A(B) \cdot P(A)}{P(B)}$$

to compute these, always use
the Law of Total Probabilities!

Ex (6) B_1, B_2, B_3 - machines to make resistors
80% of the resistors made by B_1 are accepted
90% of the resistors made by B_2 are accepted
60% of the resistors made by B_3 are accepted
(the rest are rejected)

in 1 hour, B_1 produces 3 000 resistors	$\Rightarrow$ resistors in total
B_2 produces 4 000 resistors	
B_3 produces 3 000 resistors	

a. What is the probability that the company ships an accepted resistor?

b. What is the probability that an acceptable resistor comes from machine B_3 ?

$\Rightarrow$ The space event are the three machines B_1, B_2, B_3

a. Denote by B_1 - the resistor produced by machine B_1
 B_2 - the resistor produced by machine B_2
 B_3 - the resistor produced by machine B_3
 $\{B_1, B_2, B_3\}$ - space event

A: the resistor is acceptable $\rightarrow$ the main event
 $\rightarrow$ use the Total Law of Probabilities

$$P(A) = P(B_1) \cdot P_A(B_1) + P(B_2) \cdot P_A(B_2) + P(B_3) \cdot P_A(B_3)$$

$$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow$$

$$\frac{3\,000}{10\,000} \cdot \frac{80}{100} + \frac{4\,000}{10\,000} \cdot \frac{90}{100} + \frac{3\,000}{10\,000} \cdot \frac{60}{100}$$

$$= 0,78$$

$P_{B_1}(A)$ - is the probability that the resistor is acceptable if it comes from machine B_1

$$b. P_A(B_3) = \frac{\frac{60}{100} = \frac{3000}{10000} \cdot P_{B_3}(A) \cdot P(B_3)}{P(A) = 0,78} = \frac{\frac{60}{100} \cdot \frac{3\,000}{10\,000}}{0,78} = 0,23$$

Ex ④. Assuming that W, M - equal number
 5% of M are colorblind
 0,25% of W are colorblind

a. Determine the probability of a person drawn at random is colorblind.

b. Determine the probability that having drawn a colorblind person, then it is an M.

A: the person is colorblind

Space event: $\{M, W\}$

$$P(A) = \underbrace{P(M)}_{\downarrow \frac{1}{2}} \cdot \underbrace{P_A(M)}_{\downarrow \frac{5}{100}} + \underbrace{P(W)}_{\downarrow \frac{1}{2}} \cdot \underbrace{P_A(W)}_{\downarrow \frac{0,25}{100}}$$

$$= \frac{1}{2} \cdot \frac{5}{100} + \frac{1}{2} \cdot \frac{0,25}{100} = \frac{5,25}{200} = 0.02625$$

$$P_A(M) = \frac{P_A(M) \cdot P(M)}{P(A)} = \frac{\frac{1}{20} \cdot \frac{1}{2}}{\frac{21}{200}} = \frac{1}{40} \cdot \frac{200}{21} = \frac{20}{21} = 0.952$$

Ex 8. We have 2 bags:

- One bag contains 4 mango and 2 apples
- The other contains 4 mango and 4 apples

We also have a biased coin, which can show head with the probability $P(H) = 0,6$.

if the coin shows head, we pick a random fruit from the first bag. otherwise, we pick a random fruit from the second bag.

Alice flips the coin, picks a random fruit from one of the bags and presents a mango.

What is the probability that the mango was picked from the second bag?

Space event: the two bags $\{B_1, B_2\}$

M: pick a mango $\Rightarrow$ the main event

$$P(M) = P(B_1) \cdot P_{B_1}(M) + P(B_2) \cdot P_{B_2}(M)$$

$$\downarrow \quad \quad \downarrow \quad \quad \downarrow \quad \quad \downarrow$$

$$0,6 \cdot \frac{4}{6} + 0,4 \cdot 0,5 = 0,6$$

$$P_M(B_2) = \frac{P_{B_2}(M) \cdot P(B_2)}{P(M)} = \frac{0,5 \cdot 0,4}{0,6} = \frac{1}{3}$$